Estimation of entropy rate in a fast physical random-bit generator using a chaotic semiconductor laser with intrinsic noise

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(Received 29 July 2011; published 23 January 2012)

We analyze the time for growth of bit entropy when generating nondeterministic bits using a chaotic semiconductor laser model. The mechanism for generating nondeterministic bits is modeled as a 1-bit sampling of the intensity of light output. Microscopic noise results in an ensemble of trajectories whose bit entropy increases with time. The time for the growth of bit entropy, called the memory time, depends on both noise strength and laser dynamics. It is shown that the average memory time decreases logarithmically with increase in noise strength. It is argued that the ratio of change in average memory time with change in logarithm of noise strength can be used to estimate the intrinsic dynamical entropy rate for this method of random bit generation. It is also shown that in this model the entropy rate corresponds to the maximum Lyapunov exponent.

DOI: 10.1103/PhysRevE.85.016211 PACS number(s): 05.45.Gg, 42.65.Sf, 42.55.Px, 05.45.Pq

I. INTRODUCTION

Physical random processes can be used to realize nondeterministic random number generators for use in information security and computations [1,2]. Random phenomena such as photon noise, thermal noise in resistors, and frequency jitter of oscillators have been used for physical random number generators [3–5]. Quantum noise in optical systems has been also used for physical random number generators [6–8]. However, it is difficult to realize nondeterministic generators which generate random bits at the high rates of modern digital computation and communications, which exceed gigabits per second (Gb/s).

Recently, fast physical random-bit generators using chaotic semiconductor lasers [9–19] or optical noise sources [20–22] have been demonstrated to generate random bit sequences that pass standard statistical tests of randomness at rates over Gb/s. In order to make further advances in this direction, it is important to understand the origins of the randomness properties. Moreover, from the point of view of security applications in particular, it is important to know not only that a sequence does not show statistically significant deviations from randomness, but also that the source is nondeterministic, in the sense that repeated operations of the device from the same initial state produce statistically independent sequences. In this paper we present a numerical analysis which shows how unpredictable random bits can be generated at fast rates in the chaotic laser system by amplification of intrinsic device noise, and show the relation between the inverse of memory time and the strength of the intrinsic noise.

It is well known that the output of chaotic devices can be both unpredictable as well as statistically random because they amplify effects of microscopic noise [23–25]. A sequence of bits obtained by sampling a theoretically ideal deterministic chaos system is, in principle, completely predictable if the initial state of the system is known with infinite precision. However, in any real physical device, there is always some uncertainty in the state of the device, and this will be amplified by the dynamical instability in a chaotic device so that the macroscopic state of the device after a transient time cannot be determined. This effect is used, for example, in dice and roulette to produce unpredictable results. By modeling this effect in chaotic lasers, we can theoretically estimate the entropy rate at which nondeterministic random bits can be generated. In this paper we consider the particular case of the bit extraction scheme used in Refs. [9,10], which uses 1-bit analog-to-digital conversion of the fluctuating light intensity output from a chaotic laser.

II. NUMERICAL MODEL

A set of equations for a semiconductor laser with time-delayed optical feedback from an external mirror is well known as the Lang-Kobayashi equations [26–29]. In this paper we consider the Lang-Kobayashi equations including the gain saturation effect to reproduce a similar histogram of the laser intensity distribution to that observed in experiments [27]. We also include the effects of intrinsic noise by adding stochastic noise terms. The Lang-Kobayashi equations with the gain saturation effect and stochastic noise are described as follows [27–29]:

$$\frac{dE(t)}{dt} = \frac{1}{2}\left[\frac{G_N[N(t)-N_0]}{1+\epsilon E^2(t)} - \frac{1}{\tau_p}\right]E(t) + \kappa E(t-\tau)\cos[\theta(t)] + \xi(t), \quad (1)$$

$$\frac{d\phi(t)}{dt} = \frac{\alpha}{2}\left[\frac{G_N[N(t)-N_0]}{1+\epsilon E^2(t)} - \frac{1}{\tau_p}\right] - \kappa\frac{E(t-\tau)}{E(t)}\sin[\theta(t)] + \eta(t), \quad (2)$$

$$\frac{dN(t)}{dt} = J - \frac{N(t)}{\tau_s} - \frac{G_N[N(t)-N_0]}{1+\epsilon E^2(t)}|E(t)|^2, \quad (3)$$

$$\theta = \omega\tau + \phi(t) - \phi(t-\tau), \quad (4)$$

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$$\langle \xi(t)\xi(t-\tau)\rangle = D_E\delta(\tau), \quad (5)$$

$$\langle \eta(t)\eta(t-\tau)\rangle = D_\phi\delta(\tau), \quad (6)$$

where E is the electric field amplitude, ϕ is the electric field phase, and N is the carrier density. G_N is the gain coefficient, N_0 is the carrier density at the transparency, $\kappa = (1-r_2^2)r_3/(r_2\tau_{in})$ is the optical feedback strength, r_2 is the reflectivity of the internal cavity, r_3 is the reflectivity of the external mirror, τ_{in} is the optical round-trip time in the cavity of the semiconductor laser, τ_p is the photon lifetime, τ_s is the carrier lifetime, τ is the external-cavity round-trip time, α is the linewidth enhancement factor, J is the injection current density, ϵ is the gain saturation coefficient, $\omega = 2\pi c/\lambda$ is the angular optical frequency, c is the speed of light, and λ is the optical wavelength. D_E, D_ϕ are the noise strength parameters for the electric amplitude and phase. $\langle\ \rangle$ indicates the average over time from minus to plus infinity.

Additive white Gaussian noise terms [$\xi(t)$ and $\eta(t)$] are added in each of the electric amplitude and phase equations, Eqs. (1) and (2), to model the effects of intrinsic noise [30]. For simplicity, in this paper we restrict our consideration to white Gaussian noise that affects the complex electric field in such a way that it is characterized with a single noise strength parameter D, so $D_E = D_\phi = D$ for the dimensionless Lang-Kobayashi equations [29]. The reason is to first clarify the effect of the strength of the noise in the simplest case. In numerical implementation of the above equations, the Box-Muller algorithm is used to generate white Gaussian noise from two random numbers uniformly distributed on the unit interval [31]. The random numbers were generated with the Mersenne-Twister (MT) algorithm [32], with initial conditions of MT taken from 1 to n for n noise sequences. The bandwidth of MT is determined by the inverse of the integration step ($h = 1.0\times10^{-11}$ s), which corresponds to subterahertz bandwidth.

The parameter values for the Lang-Kobayashi equations are set as follows [29]: $G_N = 8.4\times10^{-13}$ m^3 s^{-1}, $N_0 = 1.4\times10^{24}$ m^{-3}, $\tau_p = 1.927\times10^{-12}$ s, $\tau_{in} = 8.0\times10^{-12}$ s, $\tau_s = 2.04\times10^{-9}$ s, $\tau = 2.0\times10^{-9}$ s (i.e., the external cavity length of 0.3 m), $\alpha = 5.0$, $J = 1.44J_{th}$ [$J_{th} = N_{th}/\tau_s$, $N_{th} = N_0 + 1/(G_N\tau_p)$], $\kappa = 6.25$ ns^{-1}, $\lambda = 1.537\times10^{-6}$ m, and $\epsilon = 2.5\times10^{-23}$. We numerically integrated the Lang-Kobayashi equations of Eqs. (1)~(3) with these parameter values by employing the Runge-Kutta method.

III. NOISE AMPLIFICATION BY CHAOTIC DYNAMICS

We use the numerical model of Sec. II, the Lang-Kobayashi model with additive noise, to investigate the amplification of intrinsic microscopic device noise. In order to compare the value of noise strength corresponding to the noise level observed in experiments, we define the signal-to-noise ratio (SNR) of intensity as the ratio of the peak power of the center frequency to the noise floor level in the Fourier transform (FFT) of the chaotic intensity oscillation. In the numerical model, the following relationship is found:

$$SNR[\mathrm{dB}] = 5\log_{10}(D) - 45, \quad (7)$$

i.e., there is a correspondence between D of 10^0 in numerical simulation and SNR of -45 dB in experimental observation. In the experiment of Ref. [9], the SNR of -25 dB is observed, so D_E and D_ϕ of 10^4 are used in numerical simulation as a reference noise level in this section.

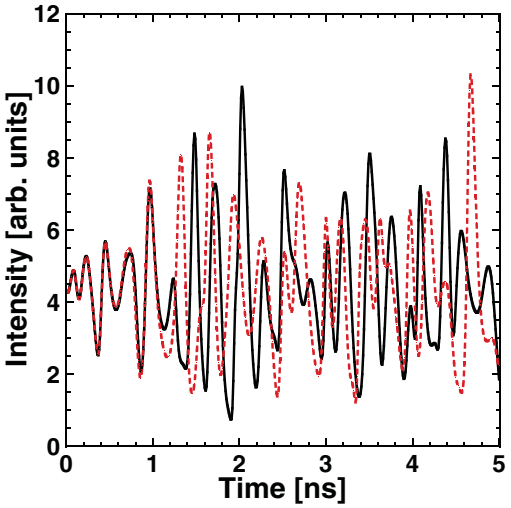


FIG. 1. (Color online) Two temporal wave forms of chaotic laser intensities starting from the same initial conditions when different noise sequences are added at time $t = 0$. The signal-to-noise ratio is –25 dB for D_E and D_ϕ of 10^4.

An ensemble of trajectories is obtained by adding different noise sequences to the same chaotic trajectory after $t = 0$. All the time series are calculated using the same parameter values. Figure 1 shows an example of two chaotic time series of the laser intensity [$I(t) = |E(t)|^2$] starting from the same initial conditions with different additive noise sequences. It is shown that the two chaotic time series are almost the same at around $t \approx 0$; however, they start separating from each other due to the effect of the different noise sequences. On the scale of this plot, the time series appear to start diverging after $1\sim2$ ns. The difference in the trajectories indicates the loss of the memory of the initial conditions due to the nonlinear amplification of the additive internal noise by chaotic dynamics.

We execute the simulation for 10^3 chaotic temporal wave forms of the laser intensity, which are generated from the same initial conditions, but are perturbed by different microscopic noise sequences after $t = 0$. The probability density function $p(t)$ (PDF, or histogram) of the laser-intensity [$I(t)$] distribution for the ensemble of the 10^3 temporal wave forms is calculated at time t. The time-dependent histogram $p(t)$ is obtained as shown in Fig. 2. For $t = 0$, the histogram has only one peak because all of the 10^3 chaotic temporal wave forms start from the same intensity value as shown in Fig. 2(a). As t is increased, the added microscopic noise is amplified by the chaotic dynamics and the standard deviation of $p(t)$ becomes larger [Figs. 2(b) and 2(c)]. Finally, $p(t)$ converges toward a certain distribution for large t [Fig. 2(d)]. The change in $p(t)$ clearly shows an increase of entropy (uncertainty) due to the amplification of the intrinsic noise by chaotic dynamics.

We confirmed that if the histogram is taken with respect to all the points along the trajectory of a single chaotic time trace without noise, the histogram will converge toward the same profile as the histogram for an ensemble of many noise-perturbed trajectories at a single time. This is expected if the chaos has the mixing property.

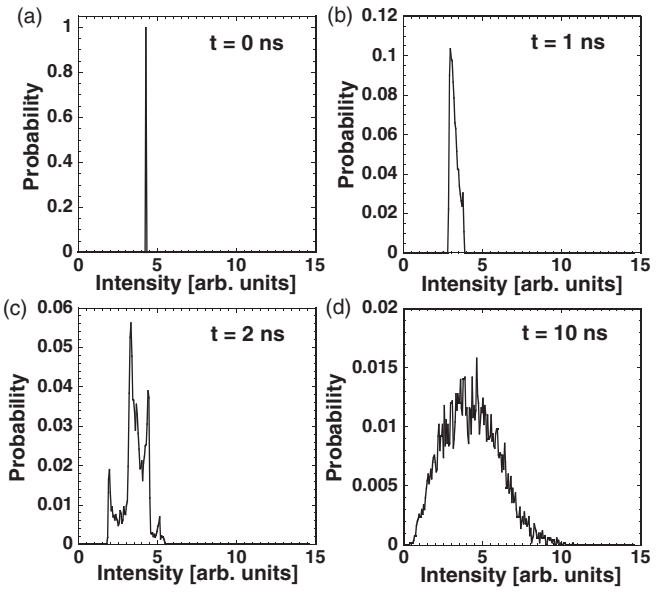


FIG. 2. Time-dependent histogram $p(t)$ of the laser-intensity distribution for different times t: (a) $t = 0$ ns, (b) $t = 1$ ns, (c) $t = 2$ ns, and (d) $t = 10$ ns, respectively. The noise strength is −25 dB.

The standard deviation $\sigma(t)$ of the time-dependent histogram $p(t)$ is calculated and plotted as a function of time. $\sigma(t)$ is normalized by the standard deviation $\sigma(\infty)$ of the final distribution of $p(\infty)$. $\sigma(t)$ is also averaged over 10^3 different initial conditions. We thus estimated the standard deviation of $p(t)$ by using the average of 10^6 temporal wave forms (10^3 different noise levels for each initial condition $\times 10^3$ different initial conditions). The initial conditions are selected on the chaotic attractor. Figure 3 shows the standard deviation ratio $\sigma(t)/\sigma(\infty)$ as a function of time. It is found that the value of $\sigma(t)/\sigma(\infty)$ increases almost monotonically and reaches ∼1 after ∼10 ns. This result is consistent with the time-dependent PDFs shown in Fig. 2.

IV. ESTIMATION OF ENTROPY RATE

Now we consider the time evolution of entropy associated with two states corresponding to a binary bit. First we calculate a temporal wave form of chaotic laser intensity. We set a threshold level to detect whether the laser intensity is above or below a specified level at fixed time intervals for extracting random bits (i.e., 1-bit analog-digital conversion) [9,10]. The threshold level is adjusted to equalize the long-time average ratio of the number of 0 and 1 bits, i.e., the probability of the occurrence of each bit 0 or 1 is as close as possible to 50%. Each point of the intensity time series is compared with the threshold level, and bit 1 is generated when the sampled value is larger than the threshold level, and bit 0 is generated otherwise. The temporal wave form is converted into a binary bit sequence in time. We execute this procedure for 10^3 temporal wave forms which are generated from the same initial conditions with different microscopic noise sequences. We then calculate the probability of bit 0 and 1 at time t, and time-dependent entropy $H(t)$ is estimated by the following equation:

$$H(t) = -\sum_{i=0}^{1} P_i(t)\log_2 P_i(t), \tag{8}$$

where $P_i(t)$ is the probability of the occurrence of bit i ($i = 0$ or 1) at the time t for an ensemble of the time series with different additive noise instances. As the width of the intensity distribution increases with time, the value of the entropy increases from zero and approaches one. Even if we know the state of the laser in the dynamical model to high precision at $t = 0$, we are unable to predict whether the wave form will correspond to a bit 0 or 1 after the entropy reaches ∼1.

Figure 4 shows an example of plots of bit entropy as a function of time for an ensemble of time series starting at two initial states (black solid and red dashed curves) at time $t = 0$.

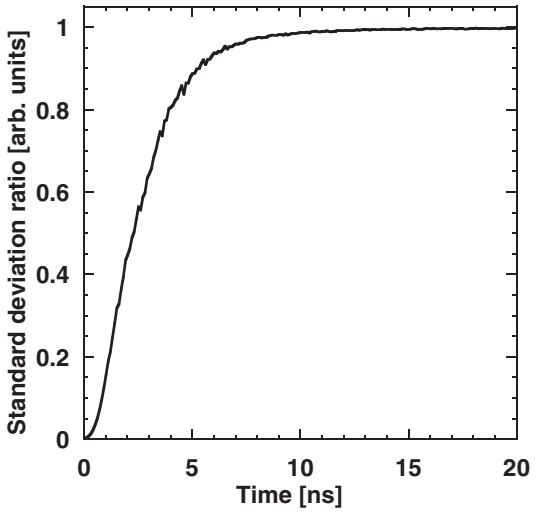


FIG. 3. Standard deviation $\sigma(t)$ of the time-dependent histogram $p(t)$, averaged over initial conditions on the chaotic attractor, as a function of time t. $\sigma(t)$ is normalized by the standard deviation $\sigma(\infty)$ of the final distribution of $p(\infty)$. The noise strength is –25 dB.

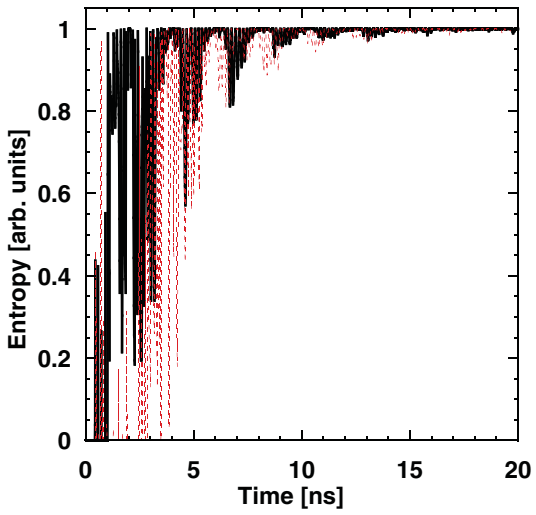


FIG. 4. (Color online) Bit entropy as a function of time for an ensemble of time series starting at two initial states [black (solid) and red (dashed) curves] at time $t = 0$. The noise strength is –25 dB.

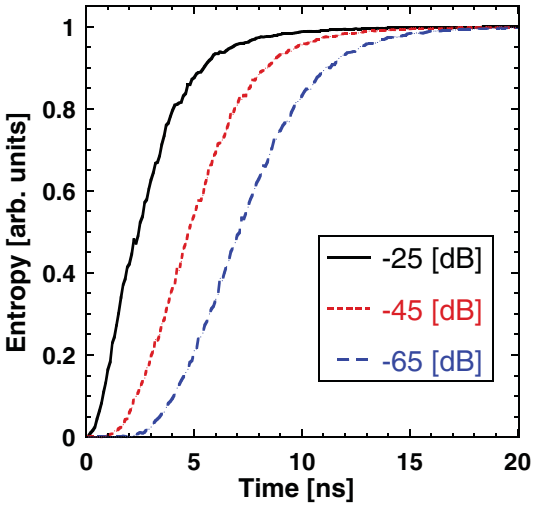


FIG. 5. (Color online) Growth of average bit entropy and its dependence on noise strength. Average bit entropy is obtained by averaging over 10^3 initial conditions and 10^3 noise instances. Black solid curve: noise strength of –25 dB ($D = 10^4$), red dotted curve: –45 dB ($D = 10^0$), blue dashed curve: –65 dB ($D = 10^{-4}$). The noise strengths are obtained from the signal-to-noise ratio in the Fourier transform of the intensity time series.

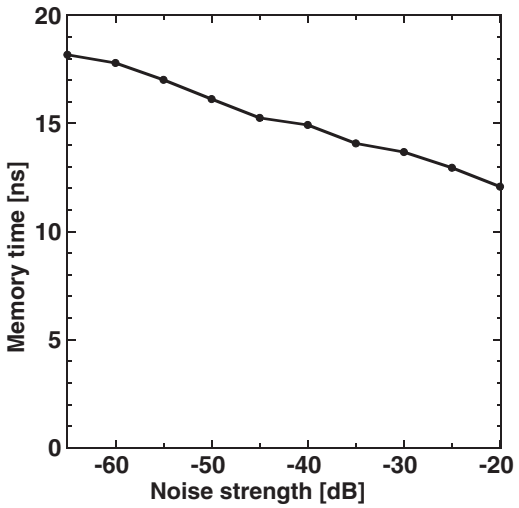


FIG. 6. Memory time as a function of the noise strength. The memory time T_m of the initial conditions is defined as the time when the time-dependent entropy $H(t)$ reaches more than 0.995. The slope of the straight line in Fig. 6 is obtained as $L = -0.136$ ns/dB.

We will define the memory time as the time required for the entropy to reach a value close to one. From Fig. 4 it can be seen that the memory time will depend on the initial state at $t = 0$, i.e., on the position on the attractor.

Figure 5 shows an example of plots of bit entropy averaged over 10^3 different initial conditions. Each plot of average entropy is obtained as a function of time for an ensemble of time series starting at exactly the same state at time $t = 0$. The average time-dependent entropy is estimated by using 10^6 temporal wave forms (10^3 different noise instances for each initial condition $\times$ 10^3 different initial conditions). Three plots are shown in Fig. 5 corresponding to three different noise strengths. It can be seen that the average entropy reaches $\sim$1 after $\sim$10 ns for the noise strength of -25 dB. It is worth noting that there is no information about the initial conditions in the bit after this time on average. As seen in Fig. 5, more time is required to converge to $H(t) \approx 1$ as the noise strength is decreased, that is, the memory time is dependent on the noise strength.

Let us specifically define the average "memory time" T_m of the initial conditions as the time when the average time-dependent entropy $H(t)$ reaches more than 0.995. This value of entropy corresponds to the pass criteria of the frequency (0/1 ratio) test for the length of 10^3 bits and significance level $\alpha = 0.01$ in the NIST Special Publication 800-22 statistical tests of randomness [33]. From Fig. 5, the memory times T_m are estimated as 13.0, 15.3, and 18.2 ns for noise strengths of –25, –45, and –65 dB, respectively. Shorter memory time is obtained for larger internal noise strength.

Figure 6 shows the result of a systematic evaluation of the dependence of memory time on noise strength. The plot is an almost straight line on the semi-logarithmic plot, indicating that the average memory time decreases in proportion to the logarithm of noise strength as noise strength is increased. This empirical relationship can be expressed as follows:

$$T_m = T_0 + L(10\log_{10} S_n), \tag{9}$$

where T_m is the memory time, T_0 is a reference time corresponding to the noise strength of 0 dB, S_n is the noise strength, and L is the slope of the straight line in Fig. 6. The logarithmic base of 10 is used in order to conveniently represent the noise strength in decibels. The value of L obtained from the plot in Fig. 6 is $L = -0.136$ ns/dB.

We found that when the threshold entropy value for the definition of the memory time is changed in the range 0.9 $\sim$ 0.995, the line is shifted downward, corresponding to a decrease in T_0; however, the slope L does not change. These results allow us to define a quantity that is independent of the noise strength, which we can call the intrinsic entropy rate. We define the entropy rate as the ratio of the change in the logarithm of the noise strength to the corresponding change in the memory time,

$$R_e = -\Delta\log_e(S_n)/\Delta T_m, \tag{10}$$

where ΔX is the change in X. The value of entropy rate R_e corresponding to the plot Fig. 6 can be obtained using Eqs. (9) and (10),

$$R_e = \frac{1}{10\log_{10} e} \cdot \frac{1}{|L|} = 1.7 \text{ ns}^{-1}. \tag{11}$$

The reason we call R_e the entropy rate is that the observed relation between memory time and noise strength can be explained if there is an initial exponential amplification of the size of the trajectory ensemble distribution at the rate R_e. We will discuss this aspect of the entropy rate further in Sec. VI.

We have also calculated the entropy rate for a laser with longer external cavity length of 3.0 m. The entropy rate obtained for the long external-cavity laser is 1.4 ns^{-1}, which is similar to the above value for the short external cavity of 0.3 m, even though the average memory time is longer for the laser with a longer external cavity due to large T_0.

V. COMPARISON OF ENTROPY RATE WITH MAXIMUM LYAPUNOV EXPONENT

To investigate the relation between the chaotic dynamics and the entropy rate, we numerically calculated the maximum Lyapunov exponent [29,34–36]. The Lang-Kobayashi equations of Eqs. (1)∼(3) are linearized around the solution, and the norm of the linearized variables over the region of $(0, \tau)$ is calculated at each step of the delay time τ. The maximum Lyapunov exponent is then obtained as the rate of the exponential growth of the norm in the logarithmic scale. For comparison, we also calculated the Kolmogorov-Sinai (KS) entropy from the sum of all the positive Lyapunov exponents [29,34].

The memory time depends on the position on the attractor, and the expansion of the perturbed trajectories at each position is expected to depend on the finite-time Lyapunov exponent. However, a simple linear relation between change in memory time and change in noise levels was obtained after averaging over initial conditions over the attractor. Hence we compare the entropy rate with the average maximum Lyapunov exponent.

Figure 7 shows the entropy rate R_e, the maximum Lyapunov exponent, and the KS entropy as a function of the optical feedback strength. The chaotic states and the corresponding Lyapunov exponents are changed as the feedback strength of the semiconductor laser is varied. The threshold value of optical intensity for bit generation was reoptimized for each feedback strength, i.e., for each value of feedback strength the threshold was adjusted so that the long-time average of the occurrence of 0 and 1 was most equal. The error bars of the entropy rate values result from the error in the linear fit for the plot of memory time versus noise strength shown in Fig. 6. We found that the entropy rate agrees well with the value of the maximum Lyapunov exponent. The correspondence is especially good for feedback strength in the range 3.3 ∼ 5 ns^{-1}. The correspondence is less exact for larger feedback strengths where fluctuation in entropy rate is larger. However, the maximum Lyapunov exponent is similar to the entropy rate over the whole regime shown in Fig. 7. On the other hand, the entropy rate is smaller than the KS entropy (i.e., the sum of all the positive Lyapunov exponents). This indicates that the 1-bit extraction method considered in this model is not optimal and there could be more efficient methods to harness the chaotic laser dynamics for random bit generation.

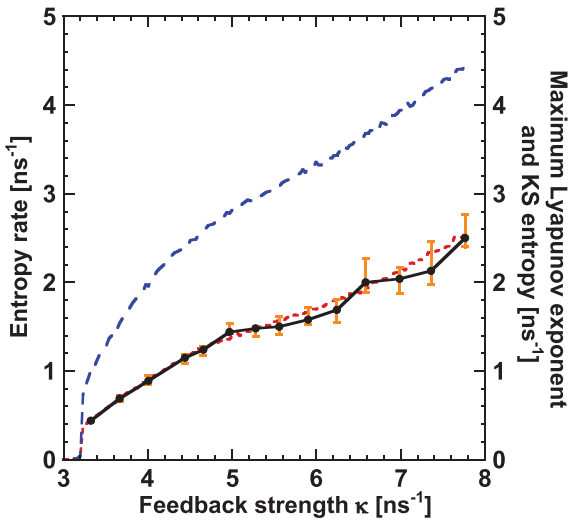


FIG. 7. (Color online) Entropy rate R_e (black solid curve with error bars), the maximum Lyapunov exponent (red dotted curve), and the Kolmogorov-Sinai entropy (blue dashed curve) as a function of the optical feedback strength. The entropy rate almost coincides with the maximum Lyapunov exponent for different chaotic states.

VI. DISCUSSION

The main focus of this work is an investigation of the growth of bit entropy due to the effect of amplification of noise by chaotic dynamics in the context of random bit generation. For this purpose we considered a particular model of a chaotic laser with ideal white noise of small amplitude, and a particular method of extracting one bit by sampling the optical intensity. According to this model, even if the state of the laser is known with high precision at some time, as the system evolves in time, the unpredictable perturbations due to the noise result in an ensemble of trajectories which spreads in volume over time. This results in an increase in the entropy of the bit extracted by sampling the intensity.

We point out that within the context of this model, random bits with high entropy could be generated at an arbitrarily large rate by sampling the noise itself with high resolution. However, when the resolution of the bit extraction is not sufficient to observe the noise, initially the entropy is zero and it is necessary to wait for some time for the entropy to increase. For a given initial state, there is an initial transient time during which the entropy fluctuates largely, but eventually it converges toward one. The average value of entropy obtained by averaging over initial states relaxes toward one monotonically. It is possible to define a characteristic memory time by specifying a threshold value of entropy close to one. The inverse of the average memory time, or the inverse of the maximum memory time (for almost all initial conditions) could be considered as the practical entropy rate. However, this practical entropy rate depends on the specific values of the noise level and the bit extraction threshold.

On the other hand, we presented evidence which supports the claim that there is a relation between the memory time and the intrinsic entropy rate of the chaotic dynamics, i.e., the rate at which entropy of high-resolution states is transferred to entropy of low-resolution states. Specifically, we showed that there was a relation between the memory time and the noise strength. This can be explained by considering the initial exponential amplification of the microscopic noise by chaotic dynamics. The volume of a small ensemble grows exponentially at a rate L given by the sum of the characteristic Lyapunov exponents, which is the intrinsic entropy rate of ensemble [37]. Hence, the difference in growth time required for two ensembles corresponding to different initial volumes V_1 and V_2 is just $(\ln V_1 - \ln V_2)/L$. This relation is similar to the relation between noise strengths and R_e [Eq. (10)]. This is the reason why we refer to R_e as an entropy rate. Note that the linear relation was obtained by averaging over initial conditions, so there is a natural correspondence with the average Lyapunov exponents by ergodic theory. The sum of the characteristic Lyapunov exponents corresponds to an

upper estimate of entropy [37]. In fact, the value of R_e was found to correspond to the maximum Lyapunov exponent. This suggests that it may be possible to obtain a larger value of R_e, and thus larger practical bit entropy rates for the same noise level, by alternative bit extraction methods.

For random bit generation using successive 1-bit sampling of the optical intensity, the time required between samples for successive bits to be random is indicated by the memory time, which is defined in this paper with respect to a specific entropy value. Using a higher entropy criterion and sampling at intervals much longer than the memory time results in bits which are more likely to be uncorrelated. Since the memory time can have a distribution of values depending on the initial state, the sampling time should be taken long enough to guarantee that the entropy is large for as many initial states as possible.

A number of methods for random bit generation, using different multibit extraction and postprocessing of bits, have been reported which give bit sequences which pass statistical tests of randomness at very high effective bit rates, up to hundreds of gigabits per second [13–16]. These results are not necessarily in contradiction with the values of the entropy rates reported in this paper. We emphasize that within the context of the model treated in this paper, an arbitrarily large rate of generating nondeterministic random bits could be achieved by directly sampling the noise component with high-resolution multibit analog-to-digital converters. On the other hand, if the bit detection threshold is larger than the noise amplitude, then it is necessary to allow the dynamics time to amplify the noise to the detection level. Hence understanding the role of noise in the bit generation system is very important for generating nondeterministic bits at high rates. The methods of this paper could be applied in the future to analyzing and better understanding the characteristics of such very high bit rate systems.

VII. CONCLUSION

We investigated the evolution of bit entropy in a model of nondeterministic physical random bit generation using a chaotic semiconductor laser with intrinsic noise. The dynamical model of the laser is based on the Lang-Kobayashi model of a semiconductor laser with delayed optical feedback. The effect of intrinsic device noise is simply modeled by white Gaussian stochastic noise added to the complex electric field. Random bit generation is modeled by sampling the optical intensity and comparing with a fixed threshold value. The entropy of bits is defined in terms of the relative number of 1 and 0 bits obtained by sampling an ensemble of trajectories with different noise instances. We investigate how the entropy of bits depends on the noise and the dynamical properties of the laser.

Starting from a known laser state corresponding to zero bit entropy, the entropy increases with time due to spread of the trajectories corresponding to different noise instances. The bit extraction threshold can be chosen so that the entropy asymptotically approaches unity. The memory time was defined as the time to reach a particular threshold value of entropy close to one. A characteristic memory time independent of the initial laser state was obtained by averaging over initial laser states. It was shown that the average memory time scales inversely with logarithm of noise strength, allowing the definition of a characteristic quantity which does not depend on the noise strength or the initial laser state. It was argued that the scaling of memory time with noise strength can be used to estimate an average entropy rate which is an intrinsic property of the dynamical laser model. Finally, we showed that for 1-bit sampling of the optical intensity in this laser model, the average entropy rate coincides with the average maximum Lyapunov exponent over a wide range of laser parameters.

ACKNOWLEDGMENTS

The authors would like to thank Apostolos Argyris, Sze-Chun Chan, Ned Corron, Daniel Gauthier, Sheng-Kwang Hwang, Lucas Illing, Ido Kanter, Laurent Larger, Fan-Yi Lin, Jia-Ming Liu, Thomas E. Murphy, Junji Ohtsubo, Damien Rontani, Rajarshi Roy, Ken Umeno, and Yun-Cai Wang for fruitful discussions and comments. A.U. gratefully acknowledges support from a Grant-in-Aid for Young Scientists from the Ministry of Education, Culture, Sports, Science and Technology in Japan and TEPCO Research Foundation. The authors would like to thank Naonori Ueda, Eisaku Maeda, Junji Yamato, Atsushi Nakamura, and Jun Muramatsu for their continuous encouragement and support.

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